

Supporting Information for

Stokes-Based 3D Polarimetric Analysis of Dipolar Near Fields

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S1. Analytical expressions of the 3D Stokes parameters

Linear electric dipole

From equation 4 of the main text, the Stokes parameters of the field emitted by an electric dipole with $\mathbf{p} = p_x \mathbf{u}_x$ are the following:

$$s_{00} = \frac{p_x^2 \{ \cos^2 \varphi [(1 - k^2 r^2 + k^4 r^4) \cos^2 \theta + 4(1 + k^2 r^2) \sin^2 \theta] + [1 - k^2 r^2 + k^4 r^4] \sin^2 \varphi \}}{16\pi^2 \varepsilon_0^2 r^6} \quad (\text{S1a})$$

$$s_{01} = -\frac{p_x^2 (1 - k^2 r^2 + k^4 r^4) \cos \theta \cos \varphi \sin \varphi}{8\pi^2 \varepsilon_0^2 r^6} \quad (\text{S1b})$$

$$s_{02} = -\frac{p_x^2 \cos^2 \varphi \sin 2\theta}{8\pi^2 \varepsilon_0^2 r^6} \quad (\text{S1c})$$

$$s_{10} = 0 \quad (\text{S1d})$$

$$s_{11} = \frac{p_x^2 (1 - k^2 r^2 + k^4 r^4) [(3 + \cos 2\theta) \cos 2\varphi - 2 \sin^2 \theta]}{64\pi^2 \varepsilon_0^2 r^6} \quad (\text{S1e})$$

$$s_{12} = \frac{p_x^2 \sin \theta \sin 2\varphi}{8\pi^2 \varepsilon_0^2 r^6} \quad (\text{S1f})$$

$$s_{20} = -\frac{p_x^2 k^3 \cos^2 \varphi \sin 2\theta}{8\pi^2 \varepsilon_0^2 r^3} \quad (\text{S1g})$$

$$s_{21} = \frac{p_x^2 k^3 \sin \theta \sin 2\varphi}{8\pi^2 \varepsilon_0^2 r^3} \quad (\text{S1h})$$

$$s_{22} = \frac{p_x^2 \{ \cos^2 \varphi [(1 - k^2 r^2 + k^4 r^4) \cos^2 \theta - 8(1 + k^2 r^2) \sin^2 \theta] + [1 - k^2 r^2 + k^4 r^4] \sin^2 \varphi \}}{16\sqrt{3}\pi^2 \varepsilon_0^2 r^6} \quad (\text{S1i})$$

Spinning electric dipole

Equation 6 of the main text represents the particular case of circular polarization $p_x = p_y = p$. A more general expression allows elliptical spinning of the dipole:

$$\mathbf{E}_{\mathbf{p},\text{el}} = \frac{e^{ikr}}{4\pi\epsilon_0 r^3} \begin{pmatrix} [-1 + kr(i + kr)] \cos \theta (p_x \cos \varphi + ip_y \sin \varphi) \\ i[-1 + kr(i + kr)] (p_y \cos \varphi + ip_x \sin \varphi) \\ 2(i + kr) \sin \theta (-ip_x \cos \varphi + p_y \sin \varphi) \end{pmatrix} \begin{pmatrix} u_\theta \\ u_\varphi \\ u_r \end{pmatrix} \quad (\text{S2})$$

From this expression, the 3D Stokes parameters are the following:

$$s_{00} = \frac{4 \sin^2 \theta (k^2 r^2 + 1) (p_x^2 \cos^2 \varphi + p_y^2 \sin^2 \varphi) + \cos^2 \theta (k^4 r^4 - k^2 r^2 + 1) (p_x^2 \cos^2 \varphi + p_y^2 \sin^2 \varphi)}{16\pi^2 \epsilon_0^2 r^6} + \frac{(k^4 r^4 - k^2 r^2 + 1) (p_x^2 \sin^2 \varphi + p_y^2 \cos^2 \varphi)}{16\pi^2 \epsilon_0^2 r^6} \quad (\text{S3a})$$

$$s_{01} = \frac{\cos \theta (k^4 r^4 - k^2 r^2 + 1) (p_x^2 - p_y^2) \sin \varphi \cos \varphi}{8\pi^2 \epsilon_0^2 r^6} \quad (\text{S3b})$$

$$s_{02} = -\frac{\sin \theta \cos \theta (p_x^2 \cos^2 \varphi + p_y^2 \sin^2 \varphi)}{4\pi^2 \epsilon_0^2 r^6} \quad (\text{S3c})$$

$$s_{10} = \frac{p_x p_y \cos \theta (k^4 r^4 - k^2 r^2 + 1)}{8\pi^2 \epsilon_0^2 r^6} \quad (\text{S3d})$$

$$s_{11} = \frac{(k^4 r^4 - k^2 r^2 + 1) [(\cos 2\theta + 3) (p_x^2 - p_y^2) \cos 2\varphi - 2 \sin^2 \theta (p_x^2 + p_y^2)]}{64\pi^2 \epsilon_0^2 r^6} \quad (\text{S3e})$$

$$s_{12} = \frac{\sin \theta [(p_x^2 - p_y^2) \sin \varphi \cos \varphi - k^3 p_x p_y r^3]}{4\pi^2 \epsilon_0^2 r^6} \quad (\text{S3f})$$

$$s_{20} = -\frac{k^3 \sin \theta \cos \theta (p_x^2 \cos^2 \varphi + p_y^2 \sin^2 \varphi)}{4\pi^2 \epsilon_0^2 r^3} \quad (\text{S3g})$$

$$s_{21} = \frac{\sin \theta [k^3 r^3 (p_x^2 - p_y^2) \sin \varphi \cos \varphi + p_x p_y]}{4\pi^2 \epsilon_0^2 r^6} \quad (\text{S3h})$$

$$s_{22} = \frac{-8 \sin^2 \theta (k^2 r^2 + 1) (p_x^2 \cos^2 \varphi + p_y^2 \sin^2 \varphi) + \cos^2 \theta (k^4 r^4 - k^2 r^2 + 1) (p_x^2 \cos^2 \varphi + p_y^2 \sin^2 \varphi)}{16\sqrt{3}\pi^2 \epsilon_0^2 r^6} + \frac{(k^4 r^4 - k^2 r^2 + 1) (p_x^2 \sin^2 \varphi + p_y^2 \cos^2 \varphi)}{16\sqrt{3}\pi^2 \epsilon_0^2 r^6} \quad (\text{S3i})$$

where further simplification can be achieved for the circular polarization case $p_x = p_y = p$:

$$s_{00} = p^2 \frac{4 \sin^2 \theta (k^2 r^2 + 1) + \cos^2 \theta (k^4 r^4 - k^2 r^2 + 1) + (k^4 r^4 - k^2 r^2 + 1)}{16 \pi^2 \varepsilon_0^2 r^6} \quad (\text{S4a})$$

$$s_{01} = 0 \quad (\text{S4b})$$

$$s_{02} = -\frac{p^2 \sin \theta \cos \theta}{4 \pi^2 \varepsilon_0^2 r^6} \quad (\text{S4c})$$

$$s_{10} = \frac{p^2 \cos \theta (k^4 r^4 - k^2 r^2 + 1)}{8 \pi^2 \varepsilon_0^2 r^6} \quad (\text{S4d})$$

$$s_{11} = -\frac{p^2 \sin^2 \theta}{16 \pi^2 \varepsilon_0^2 r^6} \quad (\text{S4e})$$

$$s_{12} = -\frac{p^2 k^3 \sin \theta}{4 \pi^2 \varepsilon_0^2 r^3} \quad (\text{S4f})$$

$$s_{20} = -\frac{p^2 k^3 \sin \theta \cos \theta}{4 \pi^2 \varepsilon_0^2 r^3} \quad (\text{S4g})$$

$$s_{21} = \frac{p^2 \sin \theta}{4 \pi^2 \varepsilon_0^2 r^6} \quad (\text{S4h})$$

$$s_{22} = p^2 \frac{-8 \sin^2 \theta (k^2 r^2 + 1) + \cos^2 \theta (k^4 r^4 - k^2 r^2 + 1) + (k^4 r^4 - k^2 r^2 + 1)}{16 \sqrt{3} \pi^2 \varepsilon_0^2 r^6} \quad (\text{S4i})$$

Linear magnetic dipole

From equation 7 of the main text, the 2D Stokes parameters of the magnetic dipole's emission with $\mathbf{m} = m_x \mathbf{u}_x$ are the following:

$$s_0 = \frac{m_x^2 k^2 (1 + k^2 r^2) [\cos^2 \theta \cos^2 \varphi + \sin^2 \varphi]}{16 \pi^2 c^2 \varepsilon_0^2 r^4} \quad (\text{S5a})$$

$$s_1 = -\frac{m_x^2 k^2 (1 + k^2 r^2) [\cos^2 \theta \cos^2 \varphi - \sin^2 \varphi]}{16 \pi^2 c^2 \varepsilon_0^2 r^4} \quad (\text{S5b})$$

$$s_2 = \frac{m_x^2 k^2 (1 + k^2 r^2) \cos \theta \sin 2\varphi}{16 \pi^2 c^2 \varepsilon_0^2 r^4} \quad (\text{S5c})$$

$$s_3 = 0 \quad (\text{S5d})$$

with s_3 being zero. Thus, a linear magnetic dipole gives only linearly polarized light. Furthermore, the identical radial profile of all non-zero Stokes parameters means that the polarimetric landscape of the magnetic dipole's emission will be identical for any radial distance.

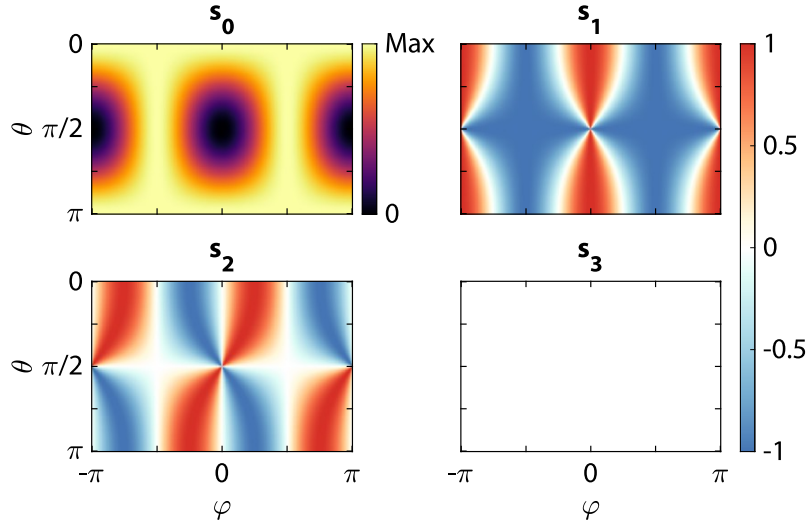


Figure S1: Normalized 2D Stokes parameters for a linear magnetic dipole, evaluated at a spherical surface at any given radial distance.

Spinning magnetic dipole

Equation 8 of the main text represents the particular case of circular polarization $m_x = m_y = m$. A more general expression allows elliptical spinning of the dipole $\mathbf{m} = m_y \mathbf{u}_y - im_x \mathbf{u}_x$:

$$\mathbf{E}_{\mathbf{m},\text{el}} = \frac{e^{ikr}}{4\pi c \varepsilon_0 r^2} k(i + kr) \begin{pmatrix} m_y \cos \varphi + im_x \sin \varphi \\ \cos \theta (m_x \cos \varphi + im_y \sin \varphi) \\ 0 \end{pmatrix} \begin{pmatrix} u_\theta \\ u_\varphi \\ u_r \end{pmatrix} \quad (\text{S6})$$

Given the two-dimensionality of this expression, polarization can be described by the conventional 2D Stokes parameters:

$$s_0 = \frac{k^2(1 + k^2 r^2) [(m_y^2 + m_x^2 \cos^2 \theta) \cos^2 \varphi + (m_x^2 + m_y^2 \cos^2 \theta) \sin^2 \varphi]}{16\pi^2 c^2 \varepsilon_0^2 r^4} \quad (\text{S7a})$$

$$s_1 = \frac{k^2(1 + k^2 r^2) [(m_y^2 - m_x^2 \cos^2 \theta) \cos^2 \varphi + (m_x^2 - m_y^2 \cos^2 \theta) \sin^2 \varphi]}{16\pi^2 c^2 \varepsilon_0^2 r^4} \quad (\text{S7b})$$

$$s_2 = \frac{k^2(1 + k^2 r^2)(m_x^2 - m_y^2) \cos \theta \sin 2\varphi}{16\pi^2 c^2 \varepsilon_0^2 r^4} \quad (\text{S7c})$$

$$s_3 = \frac{m_x m_y k^2(1 + k^2 r^2) \cos \theta}{8\pi^2 c^2 \varepsilon_0^2 r^4} \quad (\text{S7d})$$

which again can be simplified for the circular case $m_x = m_y = m$:

$$s_0 = \frac{m^2 k^2 (k^2 r^2 + 1)(1 + \cos^2 \theta)}{16\pi^2 c^2 \varepsilon_0^2 r^4} \quad (\text{S8a}) \quad s_2 = 0 \quad (\text{S8c})$$

$$s_1 = \frac{m^2 k^2 (k^2 r^2 + 1) \sin^2 \theta}{16\pi^2 c^2 \varepsilon_0^2 r^4} \quad (\text{S8b}) \quad s_3 = -\frac{m^2 k^2 (k^2 r^2 + 1) \cos \theta}{8\pi^2 c^2 \varepsilon_0^2 r^4} \quad (\text{S8d})$$

The equal radial dependence of all non-zero parameters, given by the $\frac{k^2(k^2r^2+1)}{r^4}$ fraction, means that the angular distribution of the Stokes parameters, shown below in Figure S2 will also be replicated at all radial distances. In particular, the circular spinning dipole polarization is characterized by conventional circular polarization at the poles and linear polarization (in the zenithal direction) at $\theta = \pi/2$.

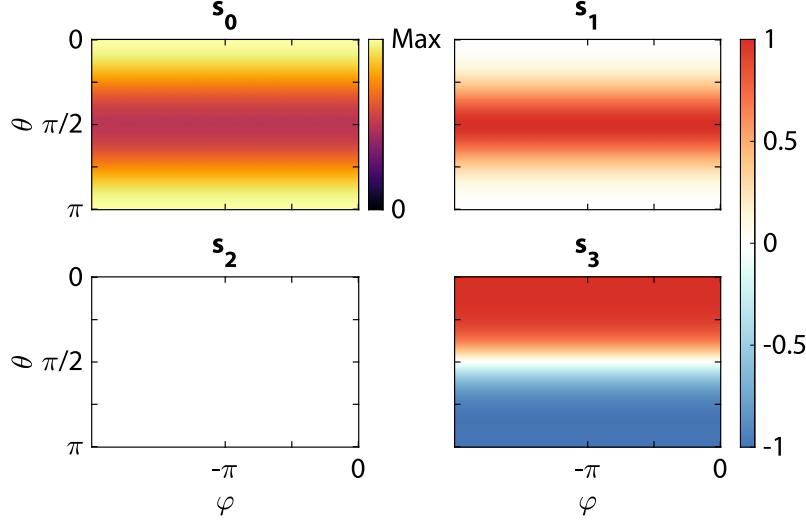


Figure S2: Normalized 2D Stokes parameters for a spinning magnetic dipole, evaluated at a spherical surface at any given radial distance.

Chiral dipole

Finally, a chiral dipole $\mathbf{p}_\xi = (p_x + ic\xi m_x)\mathbf{u}_x$ gives the following expressions for the 3D Stokes parameters:

$$s_{00} = \frac{\cos^2 \varphi \{ [c^2 p_x^2 (1 - k^2 r^2 + k^4 r^4) + \xi^2 k^2 m_x^2 r^2 (1 + k^2 r^2)] \cos^2 \theta + 4c^2 p_x^2 [1 + k^2 r^2] \sin^2 \theta \}}{16\pi^2 c^2 \varepsilon_0^2 r^6} + \frac{[c^2 p_x^2 (1 - k^2 r^2 + k^4 r^4) + \xi^2 k^2 m_x^2 r^2 (1 + k^2 r^2)] \sin^2 \varphi}{16\pi^2 c^2 \varepsilon_0^2 r^6} \quad (\text{S9a})$$

$$s_{01} = \frac{2\xi c k p_x m_x r (-\cos^2 \theta \cos^2 \varphi + \sin^2 \varphi) + [c^2 p_x^2 (-1 + k^2 r^2 - k^4 r^4) + \xi^2 k^2 m_x^2 r^2 (1 + k^2 r^2)] \cos \theta \sin 2\varphi}{16\pi^2 c^2 \varepsilon_0^2 r^6} \quad (\text{S9b})$$

$$s_{02} = \frac{p_x \cos \varphi \sin \theta [-c p_x \cos \theta \cos \varphi + \xi k m_x r (1 + k^2 r^2) \sin \varphi]}{4\pi^2 c \varepsilon_0^2 r^6} \quad (\text{S9c})$$

$$s_{10} = -\frac{\xi k^4 m_x p_x (\cos^2 \theta \cos^2 \varphi + \sin^2 \varphi)}{8\pi^2 c \varepsilon_0^2 r^2} \quad (\text{S9d})$$

$$s_{11} = \frac{\cos^2 \theta \cos^2 \varphi [c^2 p_x^2 (k^4 r^4 - k^2 r^2 + 1) - \xi^2 k^2 m_x^2 r^2 (k^2 r^2 + 1)]}{16\pi^2 c^2 \varepsilon_0^2 r^6} + \frac{\sin^2 \varphi [c^2 p_x^2 (-k^4 r^4 + k^2 r^2 - 1) + \xi^2 k^2 m_x^2 r^2 (k^2 r^2 + 1)] - 4\xi c k p_x m_x r \cos \theta \sin \varphi \cos \varphi}{16\pi^2 c^2 \varepsilon_0^2 r^6} \quad (\text{S9e})$$

$$s_{12} = \frac{p_x \cos \varphi \sin \theta [cp_x \sin \varphi + \xi k m_x r (1 + k^2 r^2) \cos \theta \cos \varphi]}{4\pi^2 c \varepsilon_0^2 r^6} \quad (\text{S9f})$$

$$s_{20} = -\frac{k^3 p_x^2 \cos^2 \varphi \sin 2\theta}{8\pi^2 \varepsilon_0^2 r^3} \quad (\text{S9g})$$

$$s_{21} = \frac{k^3 p_x^2 \cos^2 \varphi \sin 2\theta}{8\pi^2 \varepsilon_0^2 r^3} \quad (\text{S9h})$$

$$s_{22} = \frac{\cos^2 \varphi \{ [c^2 p_x^2 (1 - k^2 r^2 + k^4 r^4) + \xi^2 k^2 m_x^2 r^2 (1 + k^2 r^2)] \cos^2 \theta - 8c^2 p_x^2 [1 + k^2 r^2] \sin^2 \theta \}}{16\sqrt{3}\pi^2 c^2 \varepsilon_0^2 r^6} + \frac{[c^2 p_x^2 (1 - k^2 r^2 + k^4 r^4) + \xi^2 k^2 m_x^2 r^2 (1 + k^2 r^2)] \sin^2 \varphi}{16\sqrt{3}\pi^2 c^2 \varepsilon_0^2 r^6} \quad (\text{S9i})$$

S2. 3D parameters for $\varphi = 0$

Linear electric dipole

Equation 5 of the main text allows to extract the following 2D Stokes parameters:

$$s_0 = \frac{p_x^2 [(1 - k^2 r^2 + k^4 r^4) \cos^2 \theta + 4(1 + k^2 r^2) \sin^2 \theta]}{16\pi^2 \varepsilon_0^2 r^6} \quad (\text{S10a})$$

$$s_1 = \frac{p_x^2 [(1 - k^2 r^2 + k^4 r^4) \cos^2 \theta - 4(1 + k^2 r^2) \sin^2 \theta]}{16\pi^2 \varepsilon_0^2 r^6} \quad (\text{S10b})$$

$$s_2 = -\frac{p_x^2 \sin 2\theta}{8\pi^2 \varepsilon_0^2 r^6} \quad (\text{S10c})$$

$$s_3 = \frac{p_x^2 k^3 \sin 2\theta}{8\pi^2 \varepsilon_0^2 r^3} \quad (\text{S10d})$$

from which the $DOLP_{2D}$ and $DOCP_{2D}$ scalar metrics can be built. The individual parameters corresponding to Figure 3 in the main text are represented in Figure S3.

Chiral dipole

Taking $\varphi = 0$ from equation 10 in the main text, we obtain the following field:

$$\mathbf{E}_\xi(\varphi = 0) = \frac{e^{ikr}}{4\pi c \varepsilon_0 r^3} \begin{pmatrix} cp_x [-1 + kr(i + kr)] \cos \theta \\ \xi m_x kr (1 - ikr) \cos \theta \\ 2cp_x (1 - ikr) \sin \theta \end{pmatrix} \begin{pmatrix} u_\theta \\ u_\varphi \\ u_r \end{pmatrix} \quad (\text{S11})$$

Since the field remains 3-dimensional, the 3D formalism is needed, with parameters given by

$$s_{00} = \frac{[c^2 p_x^2 (1 - k^2 r^2 + k^4 r^4) + \xi^2 k^2 m_x^2 r^2 (1 + k^2 r^2)] \cos^2 \theta + 4c^2 p_x^2 [1 + k^2 r^2] \sin^2 \theta}{16\pi^2 c^2 \varepsilon_0^2 r^6} \quad (\text{S12a})$$

$$s_{01} = -\frac{\xi p_x m_x k \cos^2 \theta}{8\pi^2 c \varepsilon_0^2 r^5} \quad (\text{S12b})$$

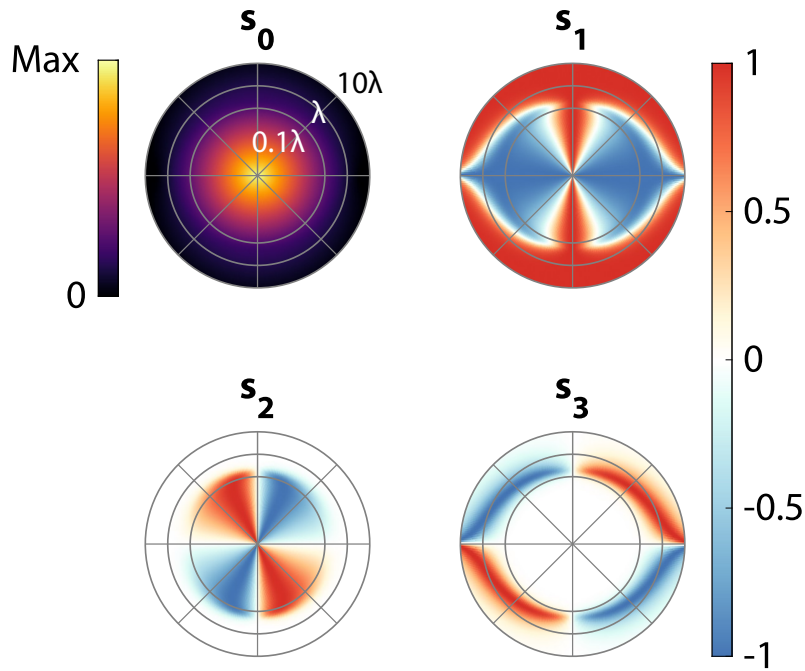


Figure S3: Normalized 2D Stokes parameters for the linear electric dipole, evaluated at the $\varphi = 0$ plane.

$$s_{02} = \frac{p_x^2 \sin \theta \cos \theta}{4\pi^2 \varepsilon_0^2 r^6} \quad (\text{S12c})$$

$$s_{10} = -\frac{\xi p_x m_x k^4 \cos^2 \theta}{8\pi^2 c \varepsilon_0^2 r^2} \quad (\text{S12d})$$

$$s_{11} = \frac{[c^2 p_x^2 (k^4 r^4 - k^2 r^2 + 1) - \xi^2 k^2 m_x^2 r^2 (k^2 r^2 + 1)] \cos^2 \theta}{16\pi^2 c^2 \varepsilon_0^2 r^6} \quad (\text{S12e})$$

$$s_{12} = \frac{\xi p_x m_x k (1 + k^2 r^2) \sin \theta \cos \theta}{4\pi^2 c \varepsilon_0^2 r^5} \quad (\text{S12f})$$

$$s_{20} = -\frac{p_x^2 k^3 \sin 2\theta}{8\pi^2 \varepsilon_0^2 r^3} \quad (\text{S12g})$$

$$s_{21} = 0 \quad (\text{S12h})$$

$$s_{22} = \frac{[c^2 p_x^2 (1 - k^2 r^2 + k^4 r^4) + \xi^2 k^2 m_x^2 r^2 (1 + k^2 r^2)] \cos^2 \theta - 8c^2 p_x^2 [1 + k^2 r^2] \sin^2 \theta}{16\sqrt{3}\pi^2 c^2 \varepsilon_0^2 r^6} \quad (\text{S12i})$$

As seen here, only s_{21} becomes zero. The normalized 3D Stokes parameters, for $\xi = 1$, are shown in Figure S4.

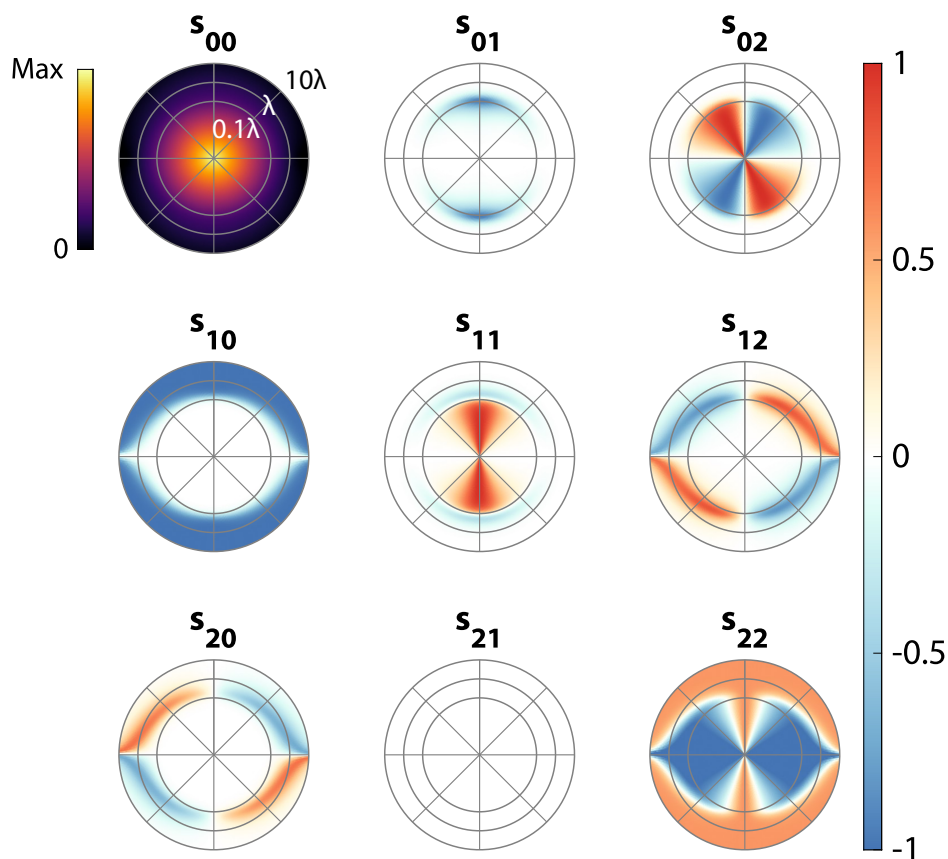


Figure S4: Normalized 3D Stokes parameters for the chiral dipole ($\xi = 1$), evaluated at the $\varphi = 0$ plane.

S3. Chiral dipole with $\xi < 1$

As noted in the main text, the case examined in Figure 5 corresponds to a maximal chirality with $\xi = 1$. While this clearly illustrates the polarimetric signatures associated with electric-magnetic dipolar coupling, realistic cases of chiral molecules and nanoparticles correspond to relatively small values of ξ . To account for such cases, Figure S5 is a replicate of Figure 5 in the main text changing the ξ value from 1 to 0.25.

As seen in the figure, a small coupling between electric and magnetic dipoles reduces most polarimetric properties of the chiral dipole to those of the linear electric dipole. Even with a moderate $\xi = 0.25$ value, electric dipolar contributions dominate and conceal all chiral features, except for small s_{10} values visible in the intermediate- and far-fields. This is further demonstrated by Figure S6, where in the $\varphi = 0$ plane many parameters become near-zero (s_{01} , s_{12}), in contrast with observations in Figure S3.

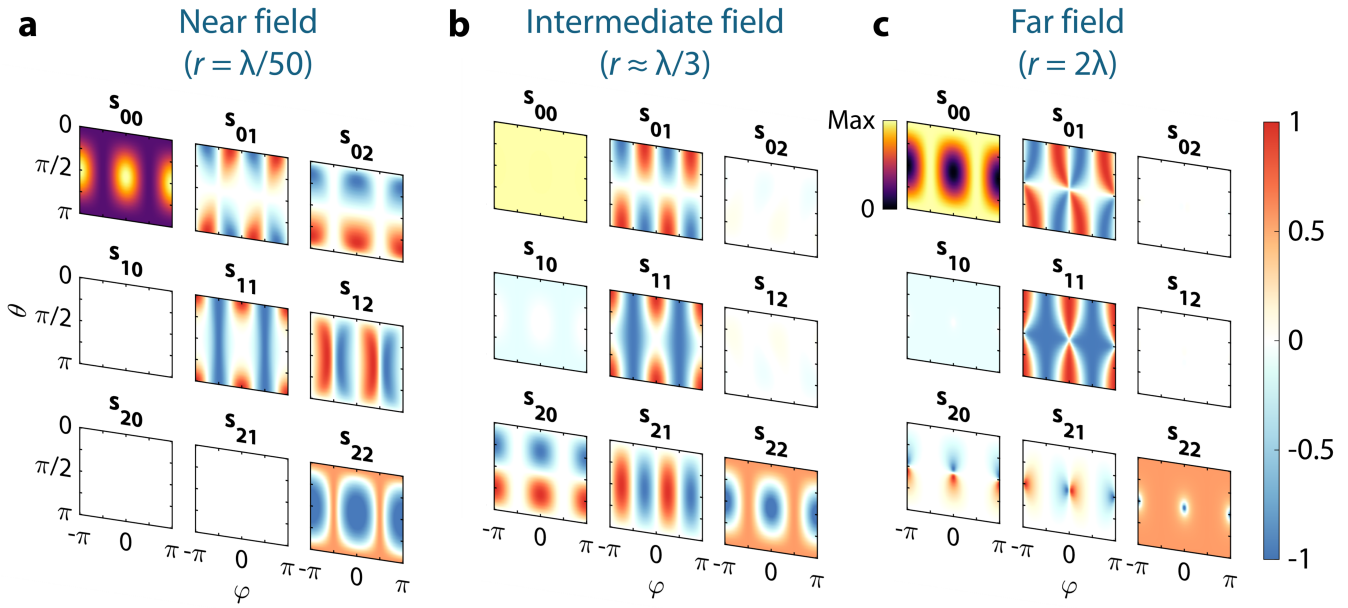


Figure S5: Normalized 3D Stokes parameters distribution over spherical surfaces enclosing the chiral dipole $\mathbf{p}_\xi = p_x + icm_x \mathbf{u}_x$ (taking $\xi = 0.25$) and radii corresponding to a) the near-field ($r = \lambda/50$), b) the intermediate-field ($r \approx \lambda/3$) and c) the far-field ($r = 2\lambda$). All parameters are normalized by s_{00} , with the exception of s_{00} itself, which denotes the total electric field intensity.

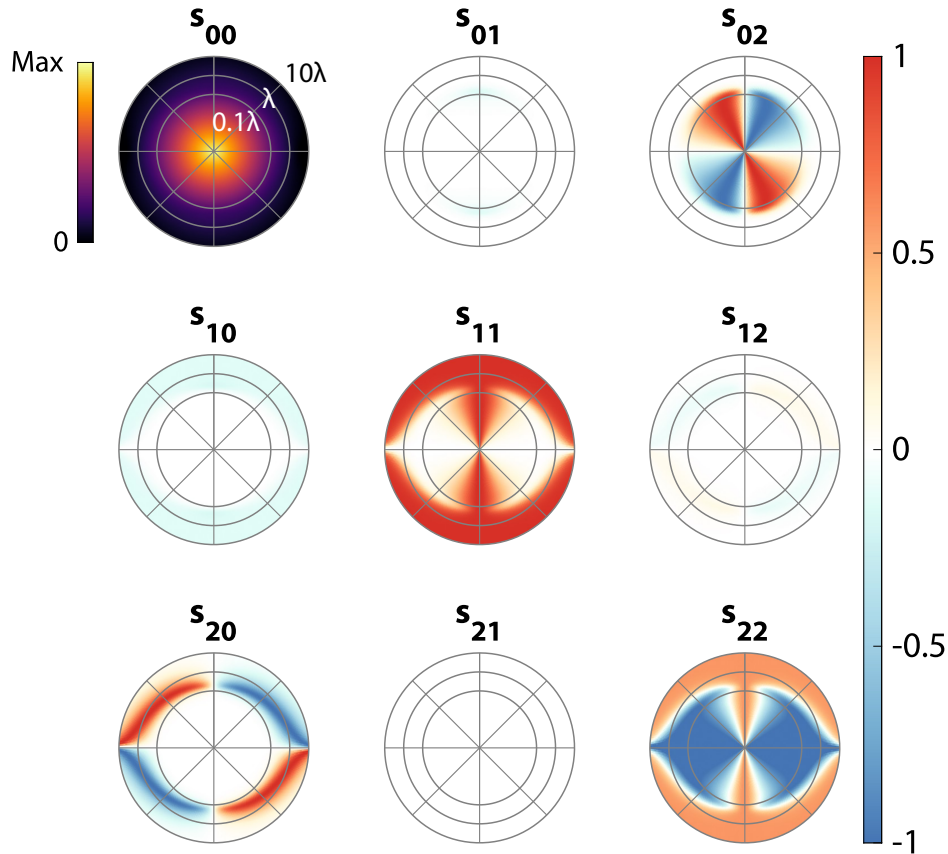


Figure S6: Normalized 3D Stokes parameters distribution for the chiral dipole $\mathbf{p}_\xi = p_x + icm_x \mathbf{u}_x$ (taking $\xi = 0.25$), evaluated at the $\varphi = 0$ plane.

S4. Identification of the dipolar intrinsic coordinate system

Unitary transformations (such as rotations) preserve the essential correlations between electromagnetic field components. Thus, a physically realizable rotation can be performed to the laboratory reference frame to accomodate the most simple view of its contents.

In the case of dipolar emitters, characterized by 2D polarization ellipses, the intrinsic system is first obtained from the electric-field spin density $\mathbf{s}_E \propto \Im(\mathbf{E}^* \times \mathbf{E})$, which matches the orientation of the polarization ellipse except for strict zero-spin $s_E = 0$ cases (for example $r \rightarrow 0$ or $r \rightarrow \infty$).

Once the spin density vector is computed, an orthonormal basis of the ellipse's tangential plane $\mathcal{T}$ can be obtained following a Gram–Schmidt process. This process yields a pair of basis vectors $\{\hat{\mathbf{t}}_1, \hat{\mathbf{t}}_2\}$ (where $\hat{\mathbf{t}}_i \cdot \mathbf{s}_E = 0$), and the coherency matrix $\mathbf{R}$ may be projected to its tangential form $\mathbf{R}_T$ as

$$\mathbf{R}_T = \mathbf{T}^\dagger \mathbf{R} \mathbf{T} \quad (\text{S13})$$

where the 3×2 matrix $\mathbf{T}$ contains the orthonormal basis of $\mathcal{T}$. A further rotation may be applied to $\mathcal{T}$ to accommodate the major axis of the ellipse along $\hat{\mathbf{t}}_1$. This rotation angle will correspond to the azimuth of the ellipse ψ , and the rotation matrix will be of the form

$$\mathbf{Q}(\psi) = \begin{pmatrix} \cos \psi & \sin \psi \\ -\sin \psi & \cos \psi \end{pmatrix} \quad (\text{S14})$$

Thus, the intrinsic coherency matrix $\mathbf{R}_O$ is given by

$$\mathbf{R}_O = \mathbf{Q}^\dagger(\psi) \mathbf{R}_T \mathbf{Q}(\psi) = \mathbf{Q}^\dagger(\psi) \mathbf{T}^\dagger \mathbf{R} \mathbf{T} \mathbf{Q}(\psi) \quad (\text{S15})$$

Finally, 2D Stokes parameters s'_i can be extracted from the intrinsic coherency matrix $\mathbf{R}_O$ as

$$s'_0 = \text{Tr}(\mathbf{R}_O) = R_{11} + R_{22} \quad (\text{S16a})$$

$$s'_1 = R_{11} - R_{22} \quad (\text{S16b})$$

$$s'_2 = 0 \quad (\text{S16c})$$

$$s'_3 = i(R_{12} - R_{21}) \quad (\text{S16d})$$

with R_{ij} being the elements of $\mathbf{R}_O$. The final rotation by ψ , allowing to align the intrinsic axes with the ellipse, mean that $s'_2 = 0$. Note that the total intensity, given by s'_0 , is invariant under any change of reference frame, since it represents the trace of the polarization matrix, and $s'_0 = \text{Tr}(\mathbf{R}) = \text{Tr}(\mathbf{R}_O)$.

S5. Spin transversality index and optical chirality

As pointed out in the discussion of Figure 6 in the main text, transversality between spin and energy flow has relatively homogeneous angular profiles. This is evidenced in Figure S7a, where the τ_S near-, intermediate- and far-field profiles of the explored dipoles are shown. Recalling that the linear dipoles have null intensity for $\varphi = \pm \frac{\pi}{2}$ and spinning dipoles at $\theta = \frac{\pi}{2}$, it becomes clear that the spin transversality index holds homogeneous values across the angular profiles.

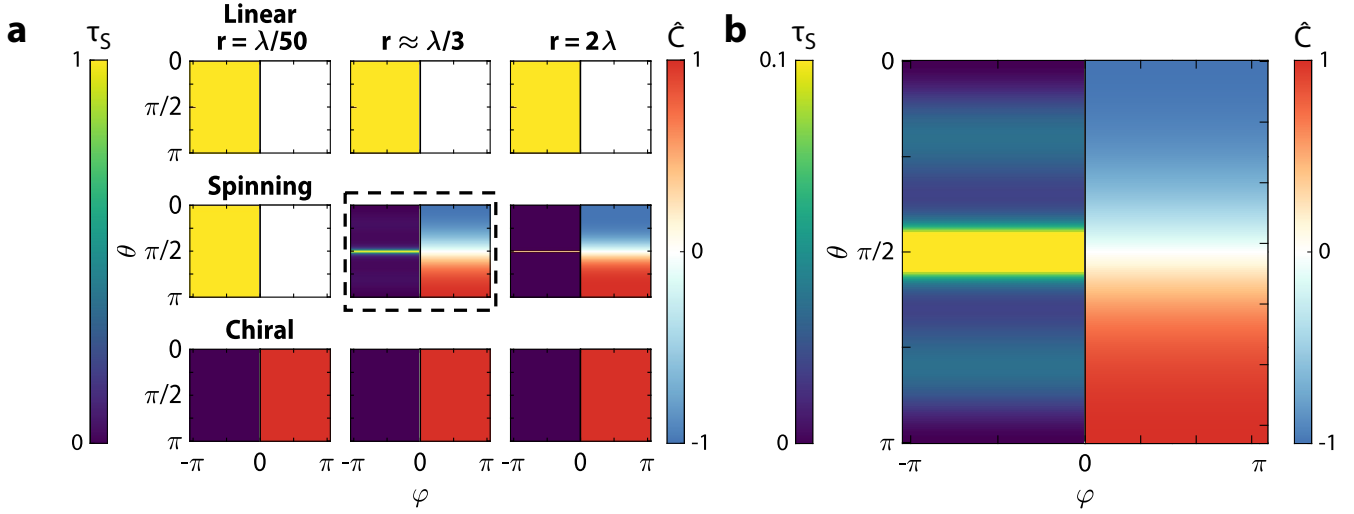


Figure S7: a) Spin transversality index τ_S (left) and normalized optical chirality density $\hat{C}$ (right) distribution over spherical surfaces enclosing the linear electric, spinning electric and chiral dipoles (taking $\xi = 1$) and radii corresponding to the near-field ($r = \lambda/50$), the intermediate-field ($r \approx \lambda/3$) and the far-field ($r = 2\lambda$). b) Close-in of the central panel in a), where the non-zero index areas around $\theta = \pi/4$ become visible.

This is also evidenced from closed-form expressions for the energy flow and spin vectors as well as the scalar optical chirality density. For a linear electric dipole, we have:

$$\mathbf{s}_E = \left[\frac{p_x^2 k^3 \sin \theta \sin \varphi \cos \varphi}{8\pi^2 \varepsilon_0 r^3}, \frac{p_x^2 k^3 \sin \theta \cos \theta \cos^2 \varphi}{8\pi^2 \varepsilon_0 r^3}, 0 \right]^T \quad (\text{S17a})$$

$$\Re(\mathbf{S}) = \left[0, 0, \frac{p_x^2 k^4 c (\cos^2 \theta \cos^2 \varphi + \sin^2 \varphi)}{32\pi^2 \varepsilon_0 r^2} \right]^T \quad (\text{S17b})$$

$$\hat{C} = \frac{\Im(\mathbf{E}^* \cdot \mathbf{H})}{|\mathbf{E}||\mathbf{H}|} = 0 \quad (\text{S17c})$$

where it is easy to see that the real part of the Poynting vector lies in the radial direction for all space, and the spin is completely transversal to it. Due to the transversality between electric and magnetic fields in all space, the optical chirality, depicted in the right half of the images in Figure S7a, is zero. Then, the spinning dipole:

$$\mathbf{s}_E = \left[\frac{p_x^2 \sin \theta}{8\pi^2 \varepsilon_0 r^6}, \frac{p_x^2 k^3 \sin \theta \cos \theta}{8\pi^2 \varepsilon_0 r^3}, \frac{p_x^2 (1 - k^2 r^2 + k^4 r^4) \cos \theta}{16\pi^2 \varepsilon_0 r^6} \right]^T \quad (\text{S18a})$$

$$\Re(\mathbf{S}) = c \left[0, \frac{p_x^2 k (1 + k^2 r^2) \sin \theta}{16\pi^2 \varepsilon_0 r^5}, -\frac{p_x^2 k^4 (1 + \xi^2) (3 + \cos 2\theta)}{64\pi^2 \varepsilon_0 r^2} \right]^T \quad (\text{S18b})$$

$$\hat{C} = -\frac{4k^3 r^3 \cos \theta}{\sqrt{[1 + k^2 r^2][3 + 2 \cos 2\theta][7 + k^2 r^2 + 3k^4 r^4 - (3 + 5k^2 r^2 - k^4 r^4) \cos 2\theta]}} \quad (\text{S18c})$$

Here, the relationship between energy propagation and spin is not so trivial, making the calculation of the spin transversality index useful. In particular, the zenithal part of the spin has a r^{-6} dependence, pointing to the dominant s_{21} observed in the near field; and the radial dependence has a dominant r^{-2} term, describing the transition to longitudinal spin as the far-field limit is

approached. Optical chirality, in this case, has a non-trivial expression which is better evaluated graphically from Figure S7a. Finally, for the chiral dipole, we have:

$$\mathbf{s}_E = \left[\frac{p_x^2 k^3 \sin \theta \sin \varphi \cos \varphi}{8\pi^2 \varepsilon_0 r^3}, \frac{p_x^2 k^3 \sin \theta \cos \theta \cos^2 \varphi}{8\pi^2 \varepsilon_0 r^3}, -\frac{p_x^2 k^4 \xi (3 + \cos 2\theta - 2 \sin \theta^2 \cos 2\varphi)}{64\pi^2 \varepsilon_0 r^2} \right]^T \quad (\text{S19a})$$

$$\Re(\mathbf{S}) = -c \left[\frac{\xi p_x^2 k^3 \sin \theta \sin \varphi \cos \varphi}{8\pi^2 \varepsilon_0 r^3}, \frac{\xi p_x^2 k^3 \sin \theta \cos \theta \cos^2 \varphi}{8\pi^2 \varepsilon_0 r^3}, -\frac{p_x^2 k^4 (1 + \xi^2) (3 + \cos 2\theta - 2 \sin \theta^2 \cos 2\varphi)}{128\pi^2 \varepsilon_0 r^2} \right]^T \quad (\text{S19b})$$

$$\hat{C} = \xi \frac{[(1 + 2k^4 r^4) \cos^2 \theta + 4(1 + k^2 r^2) \sin^2 \theta] \cos^2 \varphi + [1 + 2k^4 r^4] \sin^2 \varphi}{D} \quad (\text{S19c})$$

$$D = \sqrt{\{\cos^2 \varphi [(1 - k^2 r^2 + k^4 r^4 + \xi^2 (k^2 r^2 + k^4 r^4)) \cos^2 \theta + 4(1 + k^2 r^2) \sin^2 \theta] + [1 - k^2 r^2 + k^4 r^4 + \xi^2 (k^2 r^2 + k^4 r^4)] \sin^2 \varphi\} \cdot \sqrt{\{[(\xi^2 + k^2 r^2 + k^4 r^4 + \xi^2 (k^4 r^4 - k^2 r^2)) \cos^2 \theta \cos^2 \varphi + 4\xi^2 (1 + k^2 r^2) \sin^2 \theta \cos^2 \varphi + [\xi^2 + k^2 r^2 + k^4 r^4 + \xi^2 (k^4 r^4 - k^2 r^2)] \sin^2 \varphi\}}}} \quad (\text{S19d})$$

As seen from this equation, $\xi = \mp 1$ makes the two vectors (anti)parallel for any point in space, and thus $\tau_S(\mathbf{r}) = 0$. As the value of ξ is reduced, this condition will not be met and the angle (and therefore τ_S) will gradually increase. This is also followed by optical chirality, where $\xi = \pm 1$ makes $\hat{C} = \pm 1$.

As evidenced by these expressions, the pure linear and chiral dipoles have trivial relationships between spin transversality and chirality that hold in all space. This is not the case, however, for all dipoles. For example, weakly chiral dipoles ($\xi < 1$) will show non-trivial relationships that are geometry-dependent.

This also occurs for spinning dipoles, despite the relatively homogeneous angular profiles. Although near- and far-field approximations can give relatively straightforward relationships, the intermediate-field shows a non-zero spin transversality. Figure S7b updates the color scale of the central panel in Figure S7a to display small variations of $\tau_S \sim 0.05$ close to the chosen direction of $\theta = \frac{\pi}{4}$, which are associated with the non-zero s_{20} areas in Figure 4 of the main text. This also showcases the lack of general correlation between optical chirality and spin transversality found elsewhere, as the transverse spin areas do not seem to have an effect on the normalized optical chirality. In such situations, out of the trivial cases examined here, a full evaluation of the optical spin, considering both its electric and magnetic parts, is required.